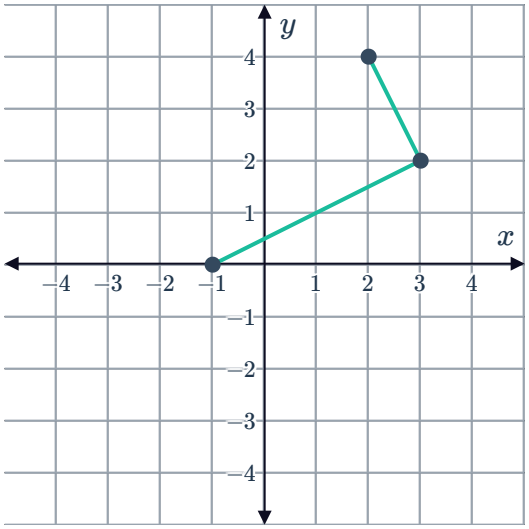
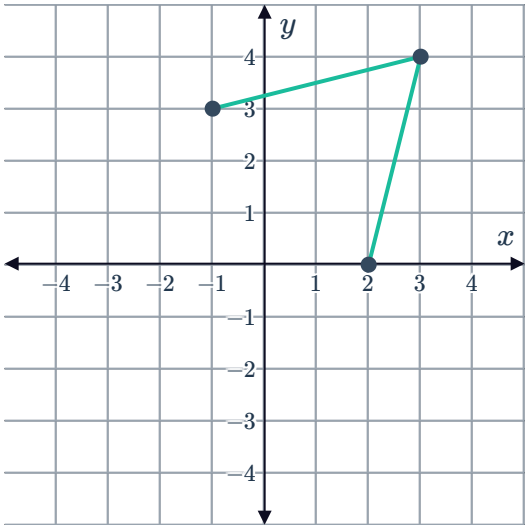
Coordinate methods with quadrilaterals

1 Consider the given diagram.
The points given represent three vertices of a parallelogram.
Determine the coordinates of the fourth vertex.



2 Consider the given diagram.
The points given represent three vertices of a rhombus.
Determine the coordinates of the fourth vertex if the missing point lies in the 3rd quadrant.

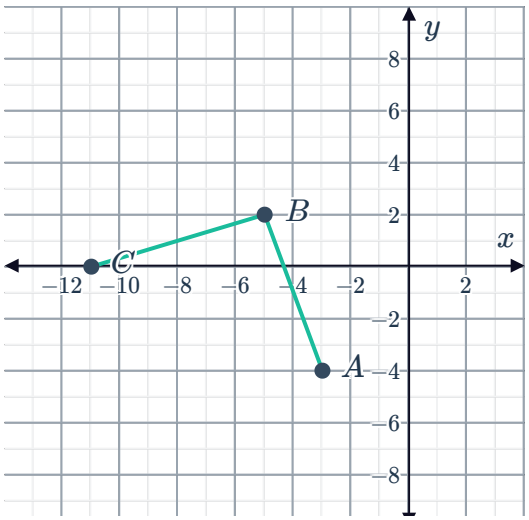


3 Consider the given diagram.
$A\,(-3,-4)$, $B\,(-5,2)$, $C\,(-11,0)$, and $D\,(p,q)$ form the four vertices of square $ABCD$, where T is the midpoint of the diagonal $\overline{AC}$.

a Find the coordinates of T.

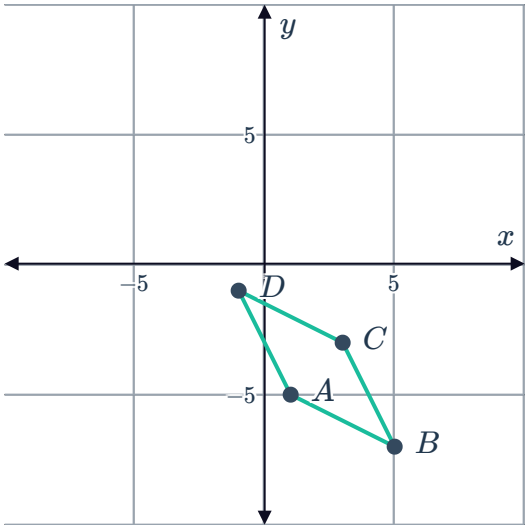
b Solve for p.

c Solve for q.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Consider the quadrilateral shown.



a Find the slope of the following:

i $\overline{AD}$ ii $\overline{AB}$

iii $\overline{BC}$ iv $\overline{DC}$

b Find the length of the following sides:

i $\overline{AB}$ ii $\overline{AD}$

c Is this quadrilateral a rhombus or just a parallelogram?

5 The four points $A\,(-8,-4)$, $B\,(-5,0)$, $C\,(0,-2)$ and $D\,(-3,-6)$ are the vertices of a quadrilateral.

a Plot the quadrilateral on a coordinate plane.

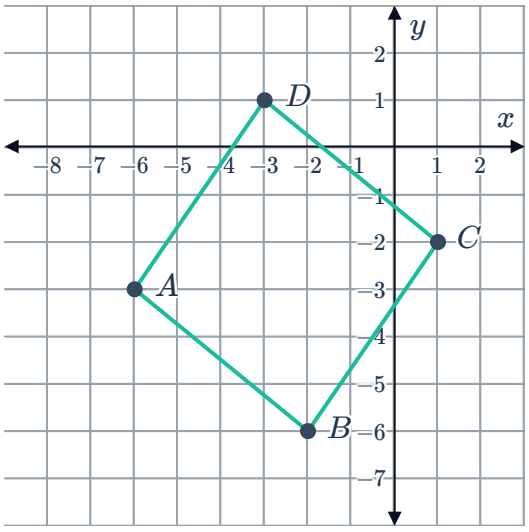
b Find the slope of the following:

i $\overline{AB}$ ii $\overline{BC}$ iii $\overline{CD}$ iv $\overline{DA}$

c Write down two pairs of segments that are parallel.

d Hence, classify the quadrilateral as precisely as possible.

6 Consider the quadrilateral shown.



a Find the slope of the following sides:

i $\overline{AB}$ ii $\overline{CD}$

iii $\overline{AD}$ iv $\overline{BC}$

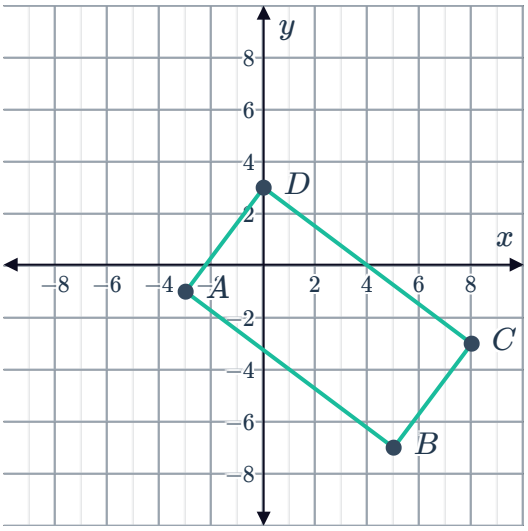
b Find the length of the following sides:

i $\overline{AB}$ ii $\overline{AD}$

c Hence, classify the quadrilateral as precisely as possible.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Consider the quadrilateral shown.



a Find the slope of the following sides:

i $\overline{AB}$ ii $\overline{CD}$

iii $\overline{AD}$ iv $\overline{BC}$

b Find the length of the following sides:

i $\overline{AB}$ ii $\overline{AD}$

c Hence, classify the quadrilateral as precisely as possible.

Additional questions

8 State the properties of each of the following triangles that can be identified with coordinate geometry:

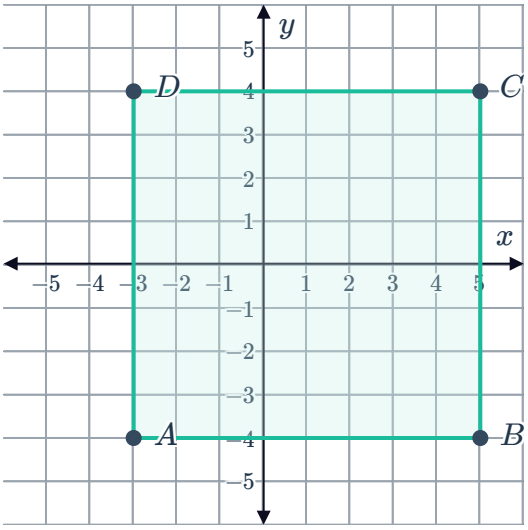
a Scalene triangles

b Isosceles triangles

c Equilateral triangles

d Right triangles

9 Consider the quadrilateral formed by the points $A\,(-3,-4)$, $B\,(5,-4)$, $C\,(5,4)$ and $D\,(-3,4)$:



a Find the length of the following sides:

i $\overline{AB}$ ii $\overline{BC}$ iii $\overline{CD}$ iv $\overline{DA}$

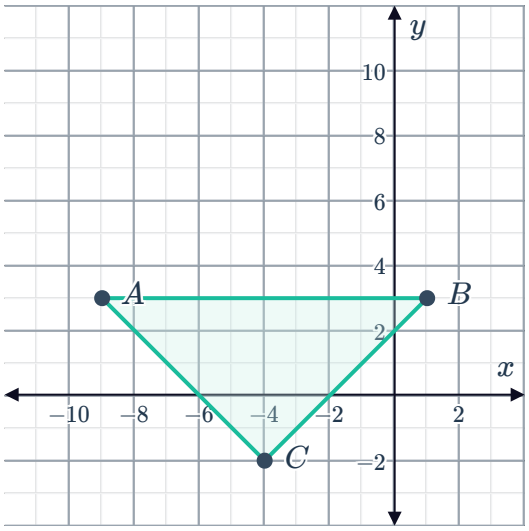
b State if all sides of the quadrilateral congruent.

c State if all consecutive sides are perpendicular.

d Now, classify the quadrilateral as precisely as possible.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Consider the triangle shown below:



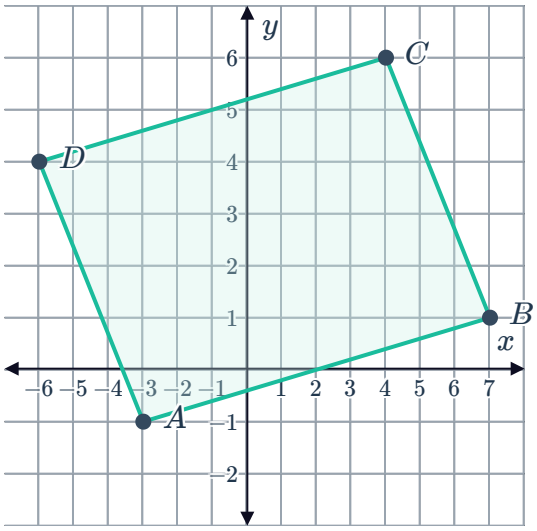
a Find the slope of the following sides:

i $\overline{AC}$ ii $\overline{BC}$

b Calculate the side lengths of all three sides.

c Classify the triangle as precisely as possible.

11 Consider the parallelogram $ABCD$:



a Find the slope of the following sides:

i $\overline{AB}$ ii $\overline{CD}$ iii $\overline{AD}$ iv $\overline{BC}$

.

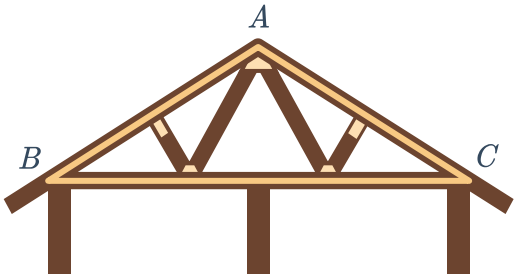
b Classify this quadrilateral as precisely as possible.

12 Isolda is designing a house using a graphing software. The outer pieces of the triangular roof trusses must form an isosceles triangle. The outer pieces of one of the triangular trusses is formed by three points: $A\,(2,2)$, $B\,(-2,0)$ and $C\,(6,0)$.

a Calculate the following lengths:

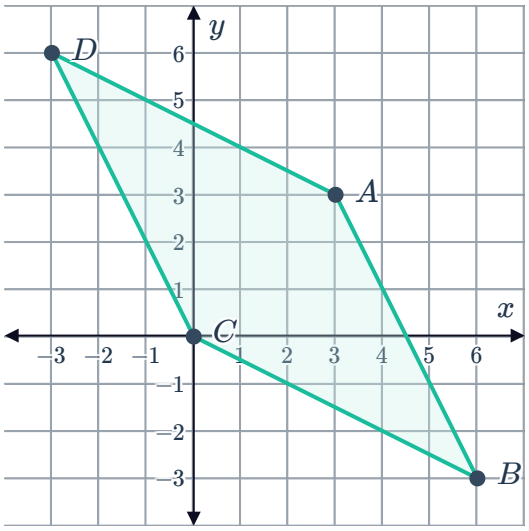
i BC ii AB iii AC

b State if this is a possible roof truss.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 Consider the quadrilateral $ABCD$:
Classify this quadrilateral as precisely as possible.



14 Consider a quadrilateral with vertices $A\,(3,4)$, $B\,(6,2)$, $C\,(5,-3)$, and $D\,(-4,3)$:

a Plot the quadrilateral on a coordinate plane.

b Classify the quadrilateral $ABCD$ as precisely as possible.

15 Consider a quadrilateral with vertices $P\,(4,-4)$, $Q\,(5,-7)$, $R\,(9,-11)$ and $S\,(0,-8)$.
Classify the quadrilateral $PQRS$ as precisely as possible.

16 A triangle consists of the points $A\,(-2,-2)$, $B\,(1,-4)$ and $C\,(5,2)$.
Classify $\triangle ABC$ as precisely as possible.

17 Consider the following lines:

- Line P: $y=-6x-4$
- Line Q: $y=\dfrac{x}{6}+6$
- Line R: $y=-6x-1$
- Line S: $y=\dfrac{x}{6}+1$

a Find the slope of each line:

i m_P **ii** m_Q **iii** m_R **iv** m_S

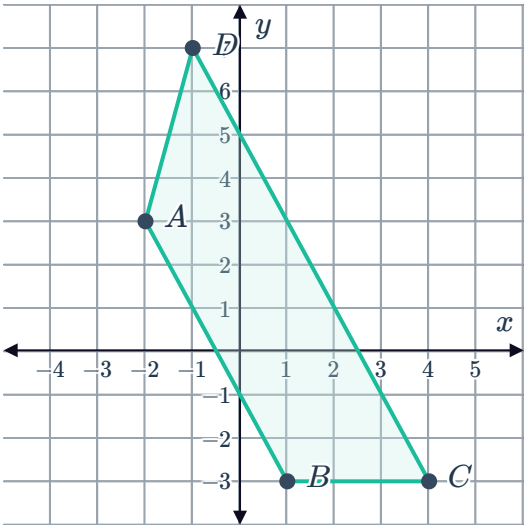
b State the type of the quadrilateral enclosed by the four lines.

18 Quadrilateral $ABCD$ has vertices $A\,(1,-2)$, $B\,(9,3)$, $C\,(7,-1)$ and $D\,(-1,-6)$:
Prove that $ABCD$ the diagonals bisect eachother and therefore that $ABCD$ is a parallelogram.

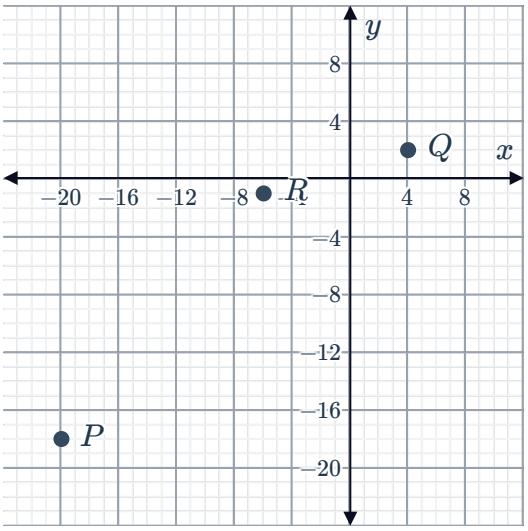
19 Quadrilateral $RSTU$ has vertices $R\,(3,0)$, $S\,(5,-3)$, $T\,(-1,-7)$, $U\,(-3,-4)$.
Classify the quadrilateral $RSTU$ as precisely as possible. Justifying your answer.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20 Prove the the given quadrilateral is a trapezoid, but not an isosceles trapezoid.



21 Consider the points $P\,(-20,-18)$, $Q\,(4,2)$ and $R\,(-6,-1)$:
Determine the coordinates of a point S such that $PQRS$ is a parallelogram.



22 Describe two ways you can show whether a quadrilateral in the coordinate plane is a square. Explain which method is more efficient.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647